
What Should a Robust Representation Preserve, Ignore, and Reject?

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Abstract

A classifier is usually trained as if $P_{\text{Train}}(X, Y) = P_{\text{Test}}(X, Y)$ and as if every test label already sits in the training set. Both break as soon as the next image has a new texture, comes from a new domain, or belongs to a new class. This assignment asks one thing about $z = g_{\theta}(x)$: what should z keep, what should it ignore, and when should the head just say it does not know? On STL-10, frozen ResNet, ViT, and CLIP heads with almost the same clean accuracy still rely on different evidence. Color barely moves the prediction; breaking the layout of the image does, especially for CLIP. On PACS, unlabeled Sketch drawings are enough: a mild pull that makes source and Sketch features look more alike raises Sketch accuracy by 8.3 points and leaves the source task intact. The same idea, done with a domain discriminator or only among Photo, Art, and Cartoon, wipes out class structure instead. SAM, which prefers a flatter source loss, is the strongest model on those three sources and the weakest on Sketch. On CIFAR, a new class that sits far from the training set (a bowl, a sunflower) is easier to reject than a near neighbor (a wolf next to dog). Making the known-class classifier a bit more accurate helps rejection a little. Mixing two known features and calling the mix “unknown” does not stand in for a real new class. What helped here was a z that still knew the object, did not lean on domain or color tricks, and could still refuse a label when the class itself was new. Code: <https://github.com/ttqureshi/ATML-PA1>.

1 Introduction

The model is $Y = f(X)$ with

$$z = g_{\theta}(x), \quad \hat{Y} = h_{\psi}(z), \quad f = g_{\theta} \circ h_{\psi}. \quad (1)$$

IID generalization is $P_{\text{Test}}(X, Y) = P_{\text{Train}}(X, Y)$. Otherwise $P_{\text{Test}} \neq P_{\text{Train}}$, which is OOD or domain generalization. A good z is supposed to be sufficient for Y , small, and stable under a small change of the input,

$$g(x + \delta x) \approx g(x). \quad (2)$$

Those two jobs pull in opposite directions. The logits are $s_k = \psi_k^{\top} z$ and the head is $\hat{y}_k = \text{softmax}(s_k)$. For a one-hot label with true class j ,

$$\mathcal{L}_{CE} = - \sum_{k=1}^K y_k \log \hat{y}_k = - \log \hat{y}_j = -s_j + \log \sum_{k=1}^K e^{s_k}. \quad (3)$$

$\mathcal{L}_{CE} \rightarrow 0$ only if the true-class logit s_j pulls away from the others. If (2) is too strong, those scores collapse. If z keeps domain-dependent coordinates z_{dd} , the head can grow s_j using a shortcut (cows on grass, camels in desert).

Those mismatches are not interchangeable. A cow can change color (a cue change). The photos can start looking different while the label rule stays the same, $P_T(x) \neq P_S(x)$ but $P_T(Y | x) = P_S(Y |$

x) (covariate shift). Or the class itself is new, $y \in \mathcal{Y}_U$ with $\mathcal{Y}_U \cap \mathcal{Y}_K = \emptyset$ (semantic shift). A K -way softmax has nowhere to put “none of these”.

The rest of the report is one question about (1): what should z keep, what should it ignore, and when should h_{ψ} refuse x instead of naming something in $\mathcal{Y}_K$? Section 3 changes one cue at a time on STL-10. Section 4 gives unlabeled Sketch from PACS and checks whether shrinking $\text{Divergence}(P_S(x), P_T(x))$ actually lowers L_T . Section 5 hides Sketch and compares ERM, source-only MMD, and SAM [Foret et al., 2021]. Section 6 asks when $u(x)$ should beat $\hat{y} = \arg \max_k z_k(x)$.

A few things were not obvious beforehand. Two heads can score the same on clean photos and still look at different cues. Making source and target features look alike does not automatically help the target. A model that is strong and flat on the sources can still fail on an unseen domain. And a better known-class classifier is not automatically a detector of new classes.

2 Notation and protocol

Head and domain. The classifier is $\hat{y}_k = \text{softmax}(\psi_k^\top z)$. In open-set scoring the same vector is written h , and $z = Wh + b$ is reserved for logits. A domain is $D_i \triangleq \{P_i(x), P_i(y | x), \mathcal{L}\}$. Source and target are D_S and D_T . ERM on a finite sample is

$$R_{\text{emp}}(f_\theta) = \frac{1}{N} \sum_{i=1}^N L(f_\theta(x_i), y_i), \quad \hat{f}_\theta = \arg \min_{\theta} R_{\text{emp}}(f_\theta). \quad (4)$$

Why source and target get aligned. Target loss is bounded by

$$L_T(f_\theta) \leq L_S(f_\theta) + \text{Divergence}(P_S(x), P_T(x)) + \min(a, b), \quad (5)$$

with

$$a = \mathbb{E}_{P_S}[|P_S(Y | x) - P_T(Y | x)|], \quad b = \mathbb{E}_{P_T}[|P_S(Y | x) - P_T(Y | x)|]. \quad (6)$$

So target error is at most source CE plus how different the inputs look plus leftover label mismatch. a and b are that leftover, averaged on the source or the target. A measure-based method adds an extra term,

$$L_{\text{Total}} = L_{CE}(D_S) + \lambda \sum_{l=l_1}^{l_2} d^2(D_S^l, D_T^l), \quad (7)$$

where d may be MMD, CORAL, or Wasserstein. MMD itself is an IPM,

$$\text{MMD}(P, Q) = \max_{f \in \mathcal{F}} |\mathbb{E}_P[f(x)] - \mathbb{E}_Q[f(x)]|. \quad (8)$$

The runs use a multi-kernel RBF estimate of (8) on the 512-d feature [Long et al., 2015]. $\min(a, b)$ is the leftover label mismatch. Matching the two clouds of z , or fooling one discriminator on z , does not force $P_S(Y | z) = P_T(Y | z)$.

Adversarial DA replaces the explicit d by a discriminator $D_{\theta_d}(z)$ that should output 1 on D_S and 0 on D_T . The feature extractor is trained on

$$L_{\text{Total}} = L_{\text{sup}}(\theta_f, \theta_y) - \lambda L_{\text{domain}}(\theta_f, \theta_d). \quad (9)$$

A gradient-reversal layer implements the minus sign: forward it is the identity, backward the domain-loss gradient is flipped,

$$\nabla_{\theta_f} L_{\text{domain}} = -\lambda \left(\frac{\partial z}{\partial \theta_f} \right) \left(\frac{\partial u}{\partial z} \right) \left(\frac{\partial L_{\text{domain}}}{\partial u} \right), \quad u = D_{\theta_d}(z). \quad (10)$$

DANN here uses (9)–(10) with λ scheduled as $\alpha(p) = 2/(1 + e^{-10p}) - 1$ [Ganin et al., 2016]. CDAN keeps the same game but feeds $\text{vec}(f \otimes p)$ instead of z , so the discriminator sees a soft class guess.

When the target is hidden. Without target samples the bound (5) becomes

$$L_T(f_\theta) \leq \sum_{i=1}^M \pi_i L_i(f_\theta) + \gamma(P_T(X), \Lambda) + \rho + \text{const.}, \quad (11)$$

where Λ is the convex hull of $\{P_i(X)\}_{i=1}^M$, γ is leftover distance from P_T to that hull, and ρ is how far the sources sit from each other. Target error is then at most the source mixture error, plus how far the target sits from the source hull, plus how spread the sources are. That leaves three things to try: lower every L_i , pull the sources closer with pairwise MMD(D_i, D_j), and hope $P_T \in \Lambda$. SAM is a different bet [Foret et al., 2021]:

$$\theta_{\text{SAM}} = \arg \min_{\theta} \max_{\|\epsilon\| \leq \rho} L(\theta + \epsilon), \quad \bar{\epsilon} = \rho \frac{\nabla_{\theta} L(\theta)}{\|\nabla_{\theta} L(\theta)\|_2}. \quad (12)$$

At a critical point the sharpness is approximately $\frac{\rho^2}{2} \lambda_{\max}(H_{\theta})$. The reported Δ_{sharp} is the first-order proxy $L(\theta + \bar{\epsilon}) - L(\theta)$ on a fixed source-val batch.

When the class set is open. Closed-set risk is $R_{\text{closed}}(\theta) = \mathbb{E}_{P_K}[\ell(f_{\theta}(x), y)]$. Open-space risk is accepted mass in unexplored space $\mathcal{O}$,

$$R_{\mathcal{O}}(f_{\theta}) = \frac{\int_{\mathcal{O}} f_{\theta}(x) dx}{\int_{S_{\mathcal{O}}} f_{\theta}(x) dx}. \quad (13)$$

The open-set objective is $\min_{\theta} R_{\mathcal{O}} + \lambda R_{\text{sup}}$. The optimal detector, if P_U were known, is the likelihood ratio

$$u^*(x) = \log \frac{P_U(x)}{P_K(x)}. \quad (14)$$

P_U is never seen, so every score used later is only a stand-in. Accept if $u(x) \leq \tau$, reject otherwise, and set τ on known validation only.

What was actually run. Seed 6304. Task 1 freezes ResNet-50, ViT-B/16, and OpenCLIP ViT-B/32 on STL-10, trains a linear h_{ψ} , and also runs CLIP zero-shot (“a photo of a {class}.”). Interventions hit a shared 224×224 RGB image before model-specific norm. Hue $+90^\circ$ keeps geometry and value. AdaIN cue conflicts use $\alpha = 1$, five pairs both ways, 200 accepted images, visual gate only (edge corr. ≥ 0.15 , pixel corr. ≤ 0.98).

Tasks 2–3 share PACS splits, a ResNet-18, frozen BN running stats, AdamW 10^{-4} , 30 epochs, patience 5, and selection by mean source-val macro-F1. Sketch labels appear only in a final eval script. Task 2 batches 8+8+8 source plus 24 unlabeled Sketch images; Task 3 drops Sketch. DAN and DAN-DG share the same multi-kernel RBF estimate of (8). Task 3 ERM is the Task 2 source-only checkpoint, not retrained. SAM uses $\rho = 0.05$ in (12).

Task 4 trains a CIFAR ResNet-18 on a 90/10 CIFAR-10 split. CIFAR-100 is evaluation only. PROSER never sees a real unknown [Zhou et al., 2021].

3 Which cues the representation uses

Clean accuracy on the 500-image STL-10 subset is already high and almost tied: 97.8 (ResNet head), 97.6 (ViT head), 98.2 (CLIP head), 96.8 (CLIP zero-shot). Appendix Table 13 has the full baseline, including the CLIP head’s strangely low mean max confidence (0.335), which is a softmax on L2-normalized 512-d features rather than a weak classifier. The gaps later are about what z actually looks at, not about a broken head.

Color is optional on this set. Grayscale removes chroma and leaves edges. Hue rotation keeps those edges and swaps the palette. Table 1 shows both edits are cheap. Accuracy drops 1.6 to 2.6 points in grayscale and 3.4 to 4.6 points after a 90° hue shift. Prediction consistency with the clean image stays above 93%. If these models care about color, they do not care much on STL-10. That matters later, because PACS Sketch is not mainly a color problem.

Shape versus texture is only defined on the images the model treats as one of the two intended classes. Ordinary photos do not say which cue won, because both cues point at the same label. Cue-conflict images break that tie (Figure 1). Shape bias and coverage are

$$SB = \frac{N_{\text{shape}}}{N_{\text{shape}} + N_{\text{texture}}}, \quad \text{coverage} = \frac{N_{\text{shape}} + N_{\text{texture}}}{N_{\text{total}}}. \quad (15)$$

Table 1: Clean accuracy and the effect of removing color, swapping hue, or destroying patch order. Consistency is agreement with the clean prediction on the same image.

Model	Clean	Grayscale		Hue rotation 90°		Patch shuffle 4 × 4	
	Acc	Acc (Δ)	Cons.	Acc (Δ)	Cons.	Acc (Δ)	Cons.
ResNet-50 (head)	97.8	96.2 (−1.6)	96.2	94.2 (−3.6)	94.8	89.6 (−8.2)	90.6
ViT-B/16 (head)	97.6	95.6 (−2.0)	95.4	94.2 (−3.4)	94.4	90.2 (−7.4)	90.6
CLIP ViT-B/32 (head)	98.2	95.8 (−2.4)	95.8	93.6 (−4.6)	93.8	81.4 (−16.8)	81.6
CLIP ViT-B/32 (zero-shot)	96.8	94.2 (−2.6)	95.2	92.6 (−4.2)	93.0	81.0 (−15.8)	81.0

Table 2: Cue-conflict counts on 200 accepted stylizations (0 rejected by the visual rule). Shape bias ignores the “other” column; coverage does not.

Model	N_{shape}	N_{texture}	N_{other}	Shape bias (%)	Coverage (%)
ResNet-50 (head)	91	57	52	61.5	74.0
ViT-B/16 (head)	119	36	45	76.8	77.5
CLIP ViT-B/32 (head)	129	17	54	88.4	73.0
CLIP ViT-B/32 (zero-shot)	117	27	56	81.2	72.0

Table 2 reports both. CLIP is the most shape-leaning (88.4% / 81.2%), then ViT (76.8%), then ResNet (61.5%). Coverage is only 72–78%. About a quarter of the predictions are some third class, so SB is a ratio on a subset, not a fact about the whole backbone. The models also differ in pretraining and size (supervised ImageNet versus image–text CLIP), so the ranking is not a clean architecture test. The safer reading is smaller: when shape and texture fight, CLIP more often stays with the content class, and ResNet more often follows the style.

Small translations barely matter. Breaking layout does. Write $T_\delta(x) = x + \delta$. f is translation-invariant if $f(T_\delta(x)) = f(x)$ and translation-equivariant if $f(T_\delta(x)) = T_\delta(f(x))$. Pooling gives a CNN some of the first. A ViT has neither, because of fixed patches and E_{pos} . The local-versus-global contrast is

$$y_i = \sum_{j \in \mathcal{L}(i)} w_{i-j} x_j \quad \text{versus} \quad y_i = \sum_{j \in G} A_{ij} x_j, \quad (16)$$

a fixed neighbourhood $\mathcal{L}(i)$ against an input-dependent sum over the whole set G . Figure 2 stays almost flat through $\delta = 32$ for every head (full numbers in Appendix Table 14), so this T_δ barely moves the prediction. The CNN and the ViT look the same on those curves. Patch shuffle is a permutation T_π of the 4×4 tiles. Ordinary self-attention is permutation-equivariant, not permutation-invariant, and a ViT with E_{pos} is not T_π -invariant. That is the test of whether z needs the layout in (16). Accuracy falls 7.4 (ViT) and 8.2 (ResNet) points, and 16.8 / 15.8 points for the two CLIP classifiers (Table 1). The t-SNE of clean versus shuffled features (Figure 3) makes the same point in representation space: class clouds smear, and CLIP’s cloud smears more. Mean cosine stability of z under the 4×4 shuffle is 0.74 (CLIP), 0.72 (ResNet), 0.65 (ViT). CLIP’s visual features care more about which patch sits where. That helps a contrastive image–text model on ordinary photos, and it hurts once the patches are scrambled.

The head and the embedding do not always vote together. Equation (2) is about z , not about $\hat{y}$. Table 3 measures it as mean $\cos(z(x), z(T(x)))$. ViT under grayscale is the clearest split: accuracy drops only 2.0 points, but cosine stability is 0.739. The linear head can keep the label while z moves. Cue-conflict stability goes the other way. ResNet and ViT features of a stylized image sit far from the content image (0.35 and 0.31), while CLIP stays closer (0.71), which matches CLIP’s higher shape bias. Zero-shot CLIP and the CLIP linear head share the visual encoder, so their intervention ranking agrees. They disagree on individual cue-conflict images (the linear head is more shape-loyal; zero-shot more often says “other”). A prompt is not the same decision as a trained ψ .

4 Aligning to an unlabeled target

PACS asks for the same seven classes under four styles (Appendix Figure 8). Source-only ERM already shows the gap (Table 4): mean source-val accuracy 93.7, Sketch 66.5. Giraffe is the class

Shape-texture cue conflicts (AdaIN)

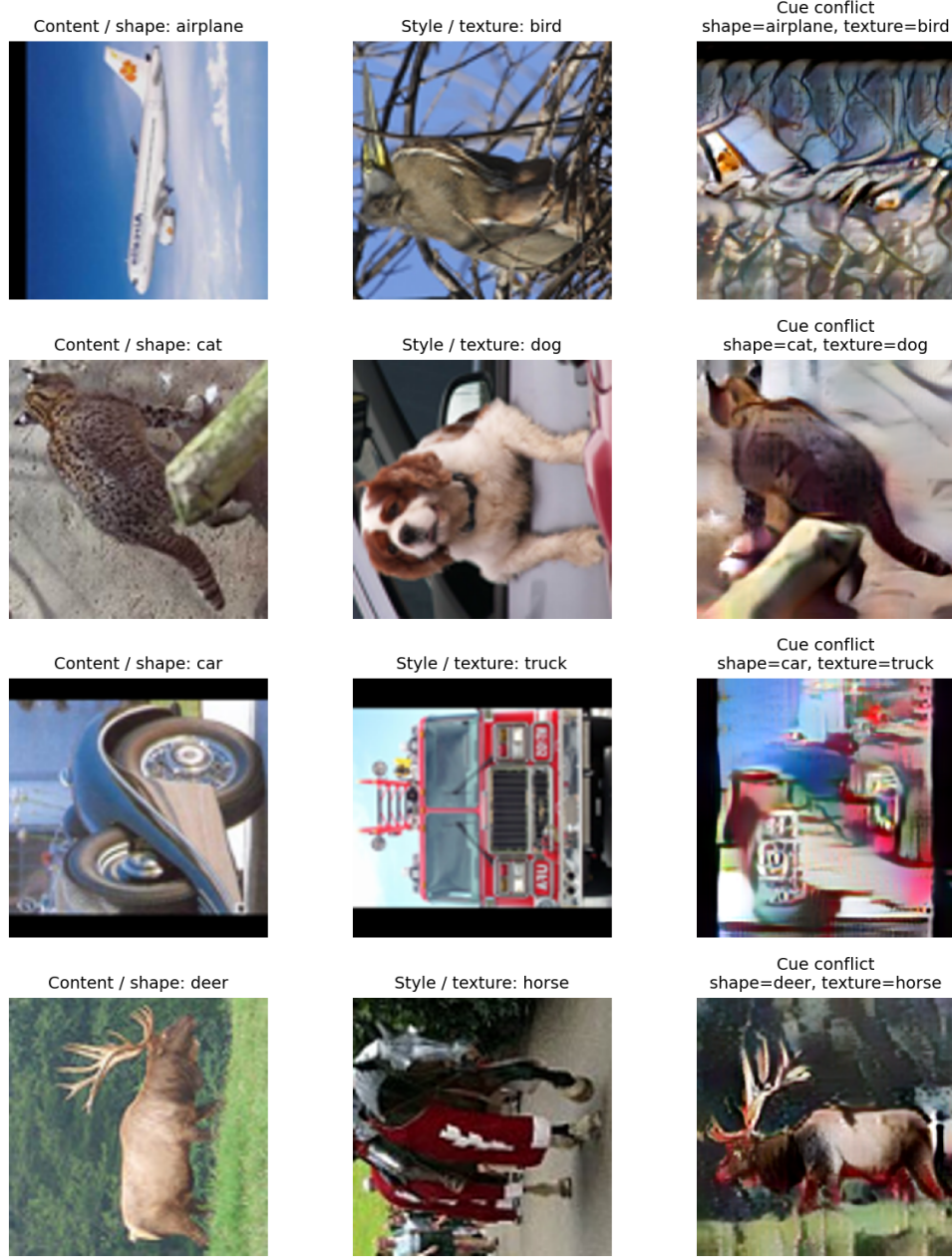


Figure 1: AdaIN cue conflicts. Content supplies shape, style supplies texture. The same image can be a shape vote, a texture vote, or neither, which is why coverage belongs next to shape bias.

the source model cannot draw (32.9); guitar is already fine (95.9). A domain probe on frozen 512-d features scores 99.6% (chance 50%). z still knows which images are Sketch.

A moderate MMD term uses that unlabeled Sketch. DAN is (7) on one layer, the 512-d feature:

$$L_{\text{Total}} = \mathcal{L}_{CE}(D_S) + \lambda_{\text{MMD}} \widehat{\text{MMD}}^2(z_S, z_T). \quad (17)$$

At $\lambda_{\text{MMD}} = 1$, Sketch rises to 74.9 (+8.3) and source F1 does not fall (Table 4). Domain separability drops from 99.6 to 95.2, not to chance. Figure 4 (center) is that compromise in feature space: source and Sketch occupy the same region. The class-colored view (Appendix Figure 10, right) still has

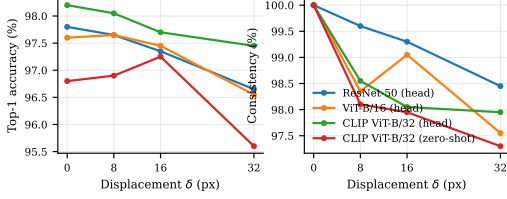


Figure 2: Accuracy and consistency under cardinal translations. All four heads stay close to clean through 32 pixels.

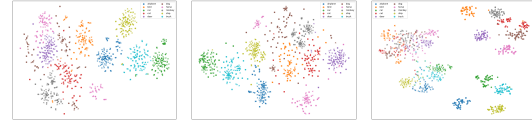


Figure 3: Clean (circles) vs. 4×4 -shuffled (crosses) features. CLIP moves the most. Grayscale embeddings: Appendix Figure 9.

Table 3: Mean cosine stability of z under each intervention. Translation uses a single 16 px right shift. CLIP zero-shot is omitted: it shares the CLIP visual features.

Backbone	Grayscale	Cue conflict	Translation (16 px)	Patch shuffle
ResNet-50 (head)	0.788	0.351	0.944	0.715
ViT-B/16 (head)	0.739	0.314	0.975	0.653
CLIP ViT-B/32 (head)	0.893	0.712	0.966	0.736

clusters. Giraffe collects most of the gain (+34.9 points, Table 8). House and person go up as well; dog slips a little. Here the middle term of (5) went down, L_T got better, and L_S in (3) stayed small. The probe is still 95.2%, so $\min(a, b)$ was not gone. It just did not get worse enough to eat the gain.

A discriminator on z can look aligned on the source loss and still destroy Sketch. DANN trains (9) through (10). Source accuracy stays 94.4 (L_S is fine) and Sketch is 27.4 (−39.1). Domain separability stays 99.5, so the middle term of (5) did not even fall. Figure 4 (right) is why: Sketch is a tight island, sources sit in rings around it. The reversed gradient moved z in a way that kept D_{θ_d} busy and wrecked s_j on Sketch. The dominant confusions are dog, horse, elephant, giraffe, and even guitar into *person* (Table 9). The min-max game can overdo it, and a discriminator that only sees z can match the wrong conditionals: $\min(a, b)$ grew. CDAN keeps (10) but replaces z by $g(x) = \text{vec}(f \otimes p)$. It is better than DANN (53.1) and worse than doing nothing. Soft p on a moving Sketch feature is a noisy class name. Animals collapse into horse instead of person (Table 9). Conditioning on the head does not fix class-conditional alignment if the head is already confused.

The class-colored DANN plot (Appendix Figure 10) is a smear. CDAN’s matching embeddings are Appendix Figure 10 (right).

Figure 5 is the same story over epochs. Source-only CE falls and source F1 rises, then training stops early. DAN’s MMD term drops while source F1 stays high. DANN and CDAN keep a large domain loss and a high source F1: the head is still solving Photo, Art, and Cartoon, while z is being pulled by a reversed gradient that Sketch never supervises.

λ_{MMD} is not “more invariance is better”. Table 5 varies λ without looking at Sketch labels. $\lambda = 0.1$ is basically source-only (Sketch 66.2, separability 99.5). $\lambda = 1$ is the setting that helped. $\lambda = 10$ lowers separability further (93.0) and also drops source F1 to 87.5; Sketch falls back to 67.6. Centroid ℓ_2 between mean source and mean Sketch features is smallest at $\lambda = 1$ (13.3), not at $\lambda = 10$ (40.4). A stronger MMD can move the means without helping the classes. If Sketch labels had been hidden, $\lambda = 1$ is still the setting I would keep: best source F1 among the three, and the first visible drop in the domain probe. Separability alone would have picked $\lambda = 10$ and thrown away the classifier. Appendix Figure 11d shows the same λ runs over epochs; Appendix Figure 11a is the $\lambda = 10$ embedding.

5 Generalizing with no target access

Equation (11) leaves three options: lower every source risk, shrink the source–source radius ρ , and hope P_T still sits in Λ . Sketch is unseen in training, selection, and the source diagnostics. The ERM row in Table 6 is the same 66.5 Sketch number as Table 4. That shared checkpoint is the fair starting point.

Table 4: PACS unsupervised DA. Source columns are validation Acc / macro-F1. Domain separability is a held-out logistic probe (chance 50%). ΔAcc is Sketch accuracy minus source-only.

Method	Source val. (Acc / F1)			Mean source		Sketch (target)			Dom. sep.
	P	A	C	Acc	F1	Acc	F1	ΔAcc	
Source-only (ERM)	97.6 / 97.1	89.5 / 88.7	94.0 / 94.1	93.7	93.3	66.5	68.3	+0.0	99.6
DAN ($\lambda_{\text{MMD}} = 1$)	97.6 / 97.1	91.2 / 91.0	93.8 / 94.5	94.2	94.2	74.9	72.4	+8.3	95.2
DANN	96.7 / 96.2	91.7 / 91.9	94.7 / 95.1	94.4	94.4	27.4	34.2	-39.1	99.5
CDAN	96.7 / 96.1	93.7 / 93.2	94.2 / 94.9	94.9	94.7	53.1	55.4	-13.4	99.0

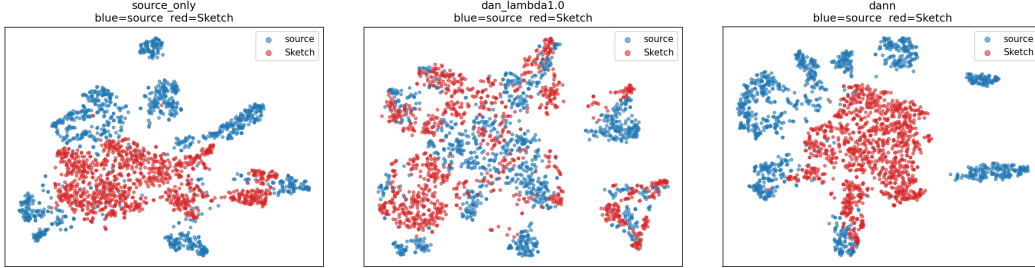


Figure 4: Learned 512-d features, domain-colored (blue source, red Sketch), same 450+450 subset. Left: ERM, Sketch is its own blob. Middle: DAN $\lambda=1$, the clouds mix. Right: DANN, Sketch is isolated again. Class-colored versions are Appendix Figure 10.

Making Photo, Art, and Cartoon look alike is not the same as making them look like Sketch. DAN-DG attacks the ρ term in (11), not the Divergence(P_S, P_T) term in (5):

$$L_{\text{Total}} = \mathcal{L}_{CE} + \lambda_{\text{DG}} \cdot \frac{1}{3} (\widehat{\text{MMD}}_{PA}^2 + \widehat{\text{MMD}}_{PC}^2 + \widehat{\text{MMD}}_{AC}^2). \quad (18)$$

The three sources are Photo, Art, Cartoon. $\lambda_{\text{DG}} = 1$. Three-way source separability falls from 89.7 to 78.1. Mean source F1 slips from 93.3 to 92.7. Sketch goes the wrong way: 63.1 (-3.4). Elephant jumps (+19.5); dog collapses (-26.0); house drops 16 points (Table 8). The sources really did get closer. That is just not the closeness Sketch needed. At $\lambda_{\text{DG}} = 10$ this is obvious (Table 7): source F1 12.2, Sketch 24.5, and the model almost always says dog. Equation (18) ate the true-class logits in (3), and $\gamma(P_T, \Lambda)$ in (11) never moved because Sketch was never in the MMD. Appendix Figure 11e shows the same thing in training: the MMD term dies, CE does not.

A flatter source loss is not a Sketch model. SAM is (12) with $\rho = 0.05$. On the sources it does its job. $\Delta_{\text{sharp}} = L(\theta + \bar{\epsilon}) - L(\theta)$ falls from 0.317 (ERM) to 0.142. Mean source F1 is the best of the three (94.9), and the worst source domain (Art) improves. Sketch is 44.6 (-22.0). Horse and guitar, two classes ERM already handled, fall by about 55 points each (Table 8). A flat region on Photo, Art, and Cartoon need not be a flat region on Sketch. Here the ranking flips: the flattest source model is the worst on the unseen domain. Mean and worst source-val F1 would have picked SAM and thrown away the ERM checkpoint that is 22 points better on Sketch. Figure 6 is the training view. SAM’s perturbed CE keeps falling; DAN-DG’s MMD falls while source F1 looks ordinary.

The two alignment setups do not help the same classes. Table 8 and Appendix Figure 11f put Task 2 and Task 3 on one grid. DAN, which sees unlabeled Sketch, mostly helps giraffe. DAN-DG, which does not, helps elephant and hurts dog. SAM barely helps anything that matters and drops horse and guitar. Table 9 is the same story as a list of mistakes: ERM mixes giraffes and horses with dogs; DANN dumps animals into person; CDAN dumps them into horse; SAM dumps horses and giraffes into dog. The shared ERM is the only fair start for asking what unlabeled Sketch added. From that start, DAN is +8.3 and DAN-DG is -3.4. That does not prove target access was the only difference. The pairing, the bandwidth sample (24 Sketch images versus three pairs of 8), and the selected epoch (14 versus 4) all change. It is still the comparison this protocol can make, and it goes one way.

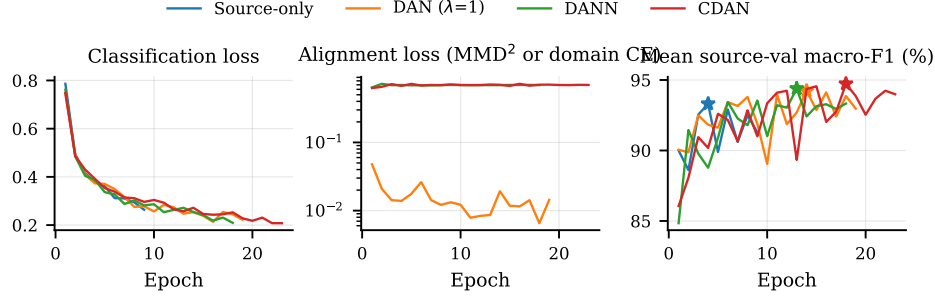


Figure 5: Task 2 training. Alignment loss is MMD^2 for DAN and domain CE for DANN / CDAN. High source F1 is cheap; Sketch is not in this plot.

Table 5: DAN alignment pressure. Centroid ℓ_2 is the distance between the mean source-val feature and the mean Sketch feature on the t-SNE diagnostic subset.

λ_{MMD}	Src Acc	Src F1	Dom. sep.	Centroid ℓ_2	Sketch Acc	Sketch F1	ΔAcc
0.1	93.9	93.9	99.5	24.1	66.2	66.5	-0.4
1	94.2	94.2	95.2	13.3	74.9	72.4	+8.3
10	87.8	87.5	93.0	40.4	67.6	67.7	+1.1

6 When the label is new

UDA and DG keep $\mathcal{Y}_K$ fixed and move $P(X)$. OSR keeps P_K and asks for a second output: reject if x looks like $\mathcal{Y}_U$. Equation (3) never sees an unknown, and far from the training points the decision regions keep going as unbounded cones. That is how R_O in (13) grows: the accepted set walks into empty space. The model looks sure. That does not mean it is right.

Four scores are four different proxies for (14). On the frozen Vanilla model (Table 10)

$$u_{\text{MSP}}(x) = 1 - \max_k P_k, \quad (19)$$

$$u_{\text{MLS}}(x) = -\max_k z_k, \quad (20)$$

$$u_{\text{energy}}(x) = -\log \sum_k e^{z_k}, \quad (21)$$

$$u_{\text{MAH}}(x) = \min_{c \in \mathcal{Y}_K} (h - \mu_c)^\top \Sigma^{-1} (h - \mu_c). \quad (22)$$

P_k is relative dominance among known classes. $\max_k z_k$ is the absolute size of the winning logit. Energy is the log-partition of all logits, so it agrees with MLS when one z_k dominates the sum. Mahalanobis is distance of h to the nearest known-class cloud, with one shared Σ . Far AUROC sits in the high 80s to low 90s; near AUROC sits near 80 for every score. Rejecting a bowl is easier than rejecting a wolf. MLS and Energy are almost the same ranking, which they should be when one logit dominates the sum. Mahalanobis is the best far detector (92.1 AUROC) and not special on near (80.3): an unknown fox can land next to dog. MSP’s histogram (Figure 7, left) is a spike at 0. Most knowns, and too many unknowns, look like $p \approx [1, 0, \dots]$. Looking sure among the K known classes is not the same as looking known. The dashed τ is the only threshold known validation is allowed to pick: about 95% of known val examples are kept, and most near unknowns are still accepted (reject 27–30%). AUROC and FPR@95TPR ask different things. Both are in the table.

A slightly better closed-set model is a slightly safer rejector here. A placeholder head is not. Table 11 holds the training axis fixed (MLS) except for the extra PROSER row. GCSC changes only the augmentation. CSA goes $94.7 \rightarrow 95.1$. Near AUROC $79.8 \rightarrow 81.5$, far $88.7 \rightarrow 90.5$, far reject at τ $49.4 \rightarrow 60.0$. In this run a slightly better known-class model plus MLS did help a bit, more on far unknowns than on near ones. That is still just a correlation. CSA rewards anything that raises known accuracy, including a texture cue that later fails on a leopard. PROSER manufactures a proxy unknown [Zhou et al., 2021]. Hidden states of two known classes are mixed,

$$h_u = \lambda h_i + (1 - \lambda) h_j, \quad \lambda \sim \text{Beta}(2, 2), \quad (23)$$

Table 6: Domain generalization on the same PACS split. Source separability is a 3-way probe (chance 33%), not the Task 2 source-versus-Sketch probe. $\Delta_{\text{sharp}} = L(\theta + \epsilon) - L(\theta)$ at $\rho = 0.05$.

Method	Source validation (Acc / F1)					Sketch (Acc / F1)	ΔAcc	Src sep.	Δ_{sharp}
	P	A	C	Mean	Worst				
ERM (Task 2 source-only)	97.6 / 97.1	89.5 / 88.7	94.0 / 94.1	93.7 / 93.3	89.5 / 88.7	66.5 / 68.3	+0.0	89.7	0.317
DAN-DG ($\lambda_{\text{DG}} = 1$)	96.1 / 95.3	88.3 / 88.4	93.4 / 94.2	92.6 / 92.7	88.3 / 88.4	63.1 / 63.5	-3.4	78.1	0.258
SAM ($\rho = 0.05$)	97.0 / 96.5	93.2 / 93.0	94.9 / 95.2	95.0 / 94.9	93.2 / 93.0	44.6 / 48.7	-22.0	88.0	0.142

Table 7: DAN-DG alignment pressure, no Sketch in the batch. $\lambda_{\text{DG}} = 10$ is a collapsed source classifier, not a stronger invariant one.

λ_{DG}	Mean src F1	Worst src F1	Src sep.	Δ_{sharp}	Sketch Acc	Sketch F1
0.1	94.2	91.8	84.4	0.370	62.2	65.8
1	92.7	88.4	78.1	0.258	63.1	63.5
10	12.2	10.5	73.1	3.684	24.5	10.0

and the extra logit z_u is trained with

$$\mathcal{L}_{\text{cls}} = \ell_{\text{CE}}([z, z_u], y_j) + \beta \ell_{\text{CE}}([z \setminus y_j, z_u], y_{K+1}). \quad (24)$$

The first term keeps the true class on top. The second asks the dummy class to win once y_j is masked. CSA drops to 94.4. MLS-AUROC is close to Vanilla and worse than GCSC at the operating point. The placeholder score, the extra thing PROSER is meant to add, is the weakest row in the table (far AUROC 82.8, far reject 35%). A point halfway between cat and truck is not a sunflower. Table 12 is a hand check of Vanilla MLS at $\tau = -5.72$. bus→truck and wolf→dog are the near-unknown cases. sunflower→bird and bowl→truck are the farther ones: a feature that splits known classes from each other can still be true of things outside $\mathcal{Y}_K$.

7 What the four studies say together

Cues and style shift. Task 1 says texture still moves ResNet more than CLIP, and that layout matters most to CLIP. PACS Sketch is a line drawing of the same classes with the style stripped off. The failures that actually show up are animal mix-ups (giraffe under ERM, person under DANN, dog under SAM), not a missing-color bug. That is only a loose match. The datasets, backbones, and training recipes are different on purpose.

Invariance is not discriminability. Appendix Figure 11g is why a domain probe is a bad success metric. In Task 2, DAN lowers source-versus-Sketch separability a little and Sketch rises; DANN barely lowers it and Sketch dies; $\lambda_{\text{MMD}} = 10$ lowers it more and Sketch comes back down. In Task 3, DAN-DG lowers the 3-way source probe and Sketch falls; $\lambda_{\text{DG}} = 10$ lowers it again and the classifier disappears. Those probes never see $\min(a, b)$. DAN kept the class clouds (Appendix Figure 10). DANN wiped them (Appendix Figure 10).

Unlabeled target data is not a small implementation choice. The shared ERM checkpoint is the only clean start. Giving DAN those Sketch images produced the only alignment run that helped. Aligning the sources to each other, with the same kernels, did not replace that. The catch is real: one seed, one $\lambda = 1$, different pairing and bandwidth, and I did not rerun DANN as a DG method. If D_T is hidden, source ERM is the honest baseline here, and SAM is a warning against trusting source validation.

Recognition is not rejection. Tasks 2 and 3 always emit a PACS label. Task 4 is the only place the model is allowed to say unknown. GCSC shows that a better z for $\mathcal{Y}_K$ can also rank P_U a bit better, mostly far away. PROSER shows that mixing known features is not the same geometry as a real unknown class. Near versus far is the same split as near OOD versus far OOD, now on labels instead of styles. Rejecting a wardrobe does not mean a wolf will not come out as dog.

What to keep, ignore, and refuse. Keep the parts that still name the object after the picture changes: roughly shape and how the parts sit together. Ignore domain identity, color shortcuts, and a style that is not the class. Do not ignore the layout of those parts, and do not “ignore” domain by

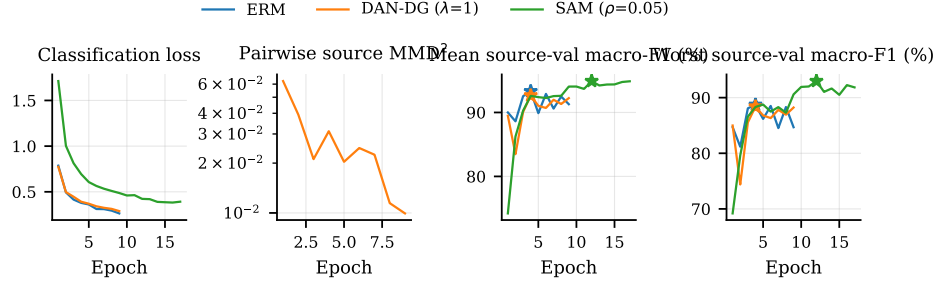


Figure 6: Task 3 training on sources only. DAN-DG logs CE and pairwise MMD. SAM logs CE at $\theta + \bar{\epsilon}$. Source F1 is the only selection signal.

Table 8: Sketch per-class accuracy and change versus ERM. n is the number of Sketch images. DAN sees unlabeled Sketch; DAN-DG and SAM do not.

Class (n)	ERM	Task 2 (sees unlabeled Sketch)			Task 3 (Sketch unseen)	
		DAN	DANN	CDAN	DAN-DG	SAM
dog (772)	59.3	56.9 (−2.5)	6.6 (−52.7)	14.2 (−45.1)	33.3 (−26.0)	66.6 (+7.3)
elephant (740)	76.6	80.3 (+3.6)	23.9 (−52.7)	26.1 (−50.5)	96.1 (+19.5)	69.1 (−7.6)
giraffe (753)	32.9	67.9 (+34.9)	20.5 (−12.5)	22.3 (−10.6)	31.9 (−1.1)	23.0 (−10.0)
guitar (608)	95.9	92.6 (−3.3)	55.9 (−40.0)	97.4 (+1.5)	87.8 (−8.1)	41.0 (−54.9)
horse (816)	68.6	73.4 (+4.8)	15.9 (−52.7)	97.2 (+28.6)	69.2 (+0.6)	12.9 (−55.8)
house (80)	81.2	100.0 (+18.8)	91.2 (+10.0)	98.8 (+17.5)	65.0 (−16.2)	80.0 (−1.2)
person (160)	83.1	96.9 (+13.7)	95.6 (+12.5)	95.0 (+11.9)	76.2 (−6.9)	84.4 (+1.2)

erasing the class clouds. Once y is outside $\mathcal{Y}_K$, invariance is the wrong tool. The head should refuse when $u(x) > \tau$, with τ set on known validation only, and it should expect near unknowns to leak through.

8 Conclusion

A good z still names the right known class after the picture changes style. It does not cheat with domain or color. And it can say unknown when the class itself is new. It does not have to be the smallest feature, and it does not have to forget every domain difference.

Unlabeled Sketch images made a mild alignment term useful. The same term, pointed at the wrong domains or turned up too far, wiped out the classes. SAM was the best model on Photo, Art, and Cartoon, and the worst on Sketch. A high softmax score does not mean the image is a known class.

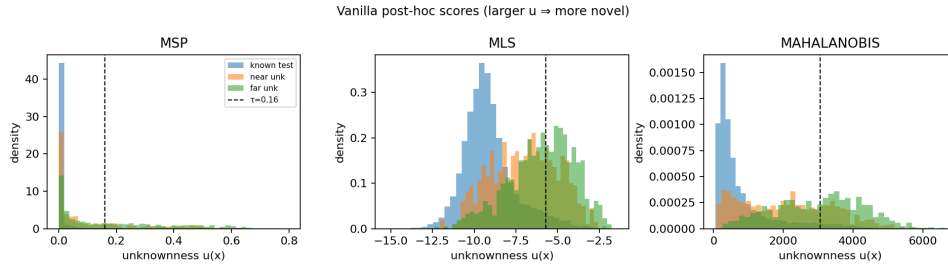
These runs use one seed and three datasets. They do not prove that GRL or SAM never work. If the target domain is hidden and the model cannot abstain, report the ERM number. Do not trust an invariance that the classes did not survive.

Table 9: Top Sketch confusions (true $\rightarrow$ predicted, count).

Model	1st	2nd	3rd
ERM	giraffe $\rightarrow$ dog (189)	horse $\rightarrow$ dog (159)	giraffe $\rightarrow$ horse (150)
DAN	dog $\rightarrow$ person (156)	dog $\rightarrow$ elephant (88)	giraffe $\rightarrow$ dog (87)
DANN	dog $\rightarrow$ person (609)	horse $\rightarrow$ person (606)	elephant $\rightarrow$ person (557)
CDAN	dog $\rightarrow$ horse (481)	elephant $\rightarrow$ horse (454)	giraffe $\rightarrow$ horse (431)
DAN-DG	dog $\rightarrow$ elephant (297)	giraffe $\rightarrow$ horse (190)	dog $\rightarrow$ horse (173)
SAM	horse $\rightarrow$ dog (573)	giraffe $\rightarrow$ dog (351)	guitar $\rightarrow$ person (278)

Table 10: Post-hoc scores on frozen Vanilla. τ is the 95th percentile of u on CIFAR-10 validation. Known acc. is CIFAR-10 test acceptance at that τ .

Score $u(x)$	AUROC (%)			At $\tau = \text{val. 95th pct. (%)}$			FPR@95TPR (%)	
	Near	Far	All	Known acc.	Near rej.	Far rej.	Near	Far
MSP	81.5	89.5	85.5	94.7	27.3	43.0	72.8	57.0
MLS	79.8	88.7	84.3	95.4	29.4	49.4	70.6	50.6
Energy	79.9	88.8	84.3	95.4	30.5	50.6	69.5	49.4
Mahalanobis	80.3	92.1	86.2	94.9	27.5	52.0	72.5	48.0

**Figure 7:** Vanilla unknownness on known test, near, and far CIFAR-100. Larger u means more novel. MSP piles everyone at zero. MLS separates far better than near. Mahalanobis stretches far unknowns away from the known peak.**Table 11:** Trained-model comparison. CSA uses only the ten known logits. The last row is PROSER’s placeholder detector, not MLS.

Model + score	AUROC (%)				At $\tau = \text{val. 95th pct. (%)}$			FPR@95TPR (%)	
	CSA	Near	Far	All	Known acc.	Near rej.	Far rej.	Near	Far
Vanilla + MLS	94.7	79.8	88.7	84.3	95.4	29.4	49.4	70.6	50.6
GCSC + MLS	95.1	81.5	90.5	86.0	94.8	34.8	60.0	65.2	40.0
PROSER + MLS	94.4	81.6	89.2	85.4	95.7	28.7	48.2	71.2	51.7
PROSER + PROSER placeholder	94.4	77.7	82.8	80.2	94.9	28.0	35.0	72.0	65.0

Table 12: False accepts of Vanilla MLS at the val-calibrated τ . Near failures are neighbor classes. Far failures are not.

Group	CIFAR-100 class	Predicted	$u_{\text{MLS}}(x)$	τ
Near	bus	truck	-12.04	-5.72
Near	wolf	dog	-12.02	-5.72
Near	bus	truck	-11.95	-5.72
Far	sunflower	bird	-12.38	-5.72
Far	bottle	cat	-11.84	-5.72
Far	bowl	truck	-11.15	-5.72

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A Extra tables and learned-feature plots

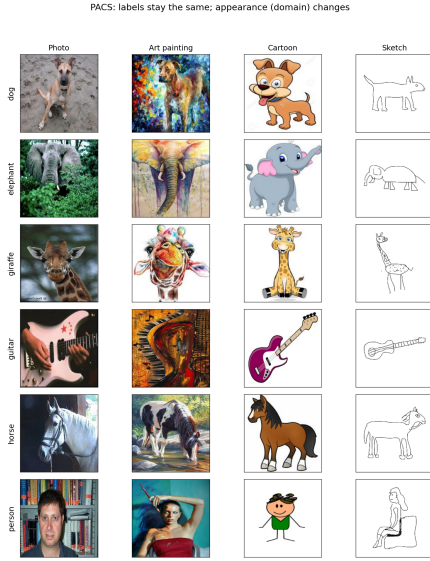


Figure 8: PACS classes across Photo, Art, Cartoon, and Sketch.

Table 13: Clean STL-10 subset (500 images). CLIP-head confidence is low because features are L2-normalized.

Model	Top-1 (%)	Macro-F1 (%)	Mean max conf.
ResNet-50 (head)	97.8	97.8	0.965
ViT-B/16 (head)	97.6	97.6	0.964
CLIP ViT-B/32 (head)	98.2	98.2	0.335
CLIP ViT-B/32 (zero-shot)	96.8	96.8	0.943

Table 14: Translation accuracy / consistency, four-direction mean.

Model	$\delta = 0$	$\delta = 8$	$\delta = 16$	$\delta = 32$
ResNet-50 (head)	97.8 / 100.0	97.6 / 99.6	97.3 / 99.3	96.6 / 98.4
ViT-B/16 (head)	97.6 / 100.0	97.6 / 98.4	97.4 / 99.0	96.5 / 97.5
CLIP ViT-B/32 (head)	98.2 / 100.0	98.0 / 98.6	97.7 / 98.0	97.5 / 98.0
CLIP ViT-B/32 (zero-shot)	96.8 / 100.0	96.9 / 98.1	97.2 / 97.9	95.6 / 97.3

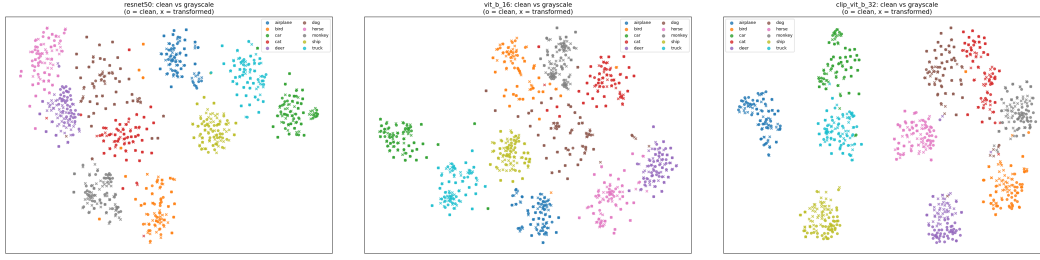


Figure 9: Clean versus grayscale features. Class geometry mostly stays; the clouds still shift.

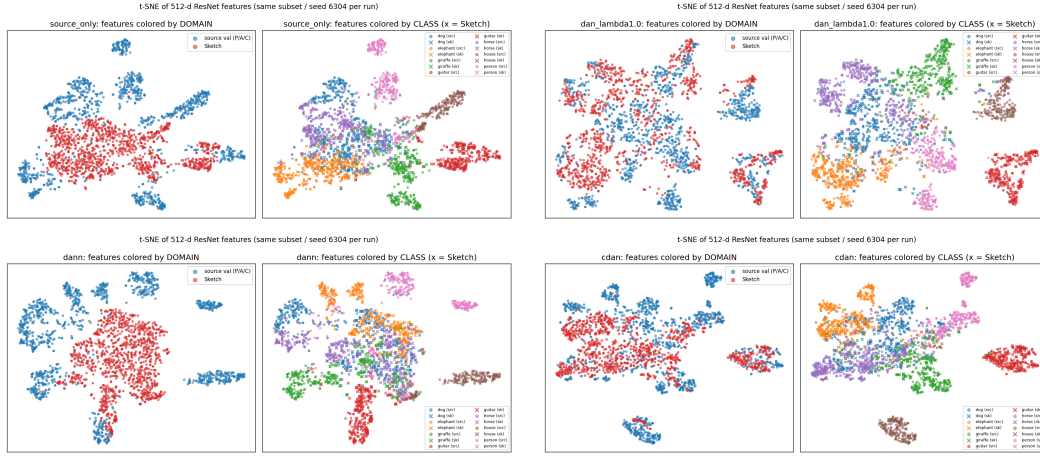
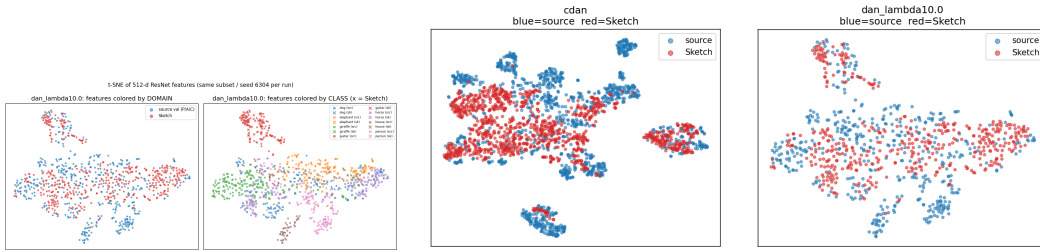


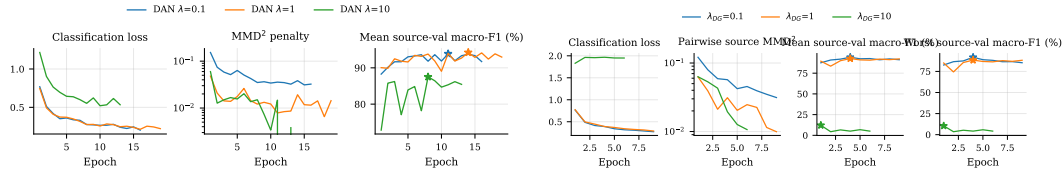
Figure 10: Class-colored 512-d features. Top: source-only and DAN $\lambda_{\text{MMD}} = 1$. Bottom: DANN and CDAN. Crosses are Sketch. DAN mixes domains and keeps class clouds; DANN isolates Sketch.



(a) DAN $\lambda_{\text{MMD}} = 10$.

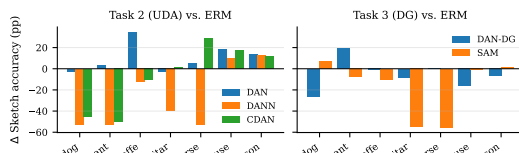
(b) CDAN, by domain.

(c) DAN $\lambda = 10$, by domain.

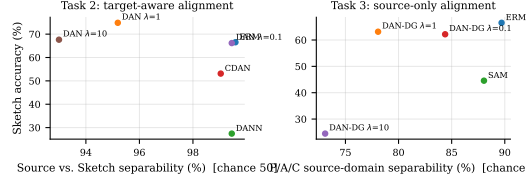


(d) DAN λ training.

(e) DAN-DG λ_{DG} training.



(f) Per-class Sketch change versus ERM.



(g) Domain probe versus Sketch accuracy.

Figure 11: Leftover λ embeddings, training curves, per-class Sketch change, and the domain-probe plot.